



Alberta Public Health Disease Management Guidelines

Yellow Fever



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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Case Definition

Confirmed Case

Clinical illness^(A) with laboratory confirmation of infection:

- Isolation of yellow fever virus from an appropriate clinical specimen

OR

- Detection of yellow fever antigen or nucleic acid (e.g., PCR) in an appropriate clinical specimen (e.g., body fluids or tissue)^(B)

OR

- Seroconversion or significant (i.e., fourfold or greater) rise in HAI titre to the yellow fever virus in the absence of yellow fever vaccination

OR

- A single elevated yellow fever IgM^(C) in the absence of yellow fever vaccination within the previous two months and in the absence of other elevated flavivirus titres

Probable Case

The following probable case definition is provided as a guideline to assist with case finding and public health management, and should not be reported to Alberta Health.

Clinical illness^(A) with laboratory evidence of infection:

- A stable elevated HAI titre to yellow fever virus with no other known cause

AND

- Cross-reactive serologic reactions to other flaviviruses has been excluded

AND

- The patient does not have a history of yellow fever vaccination

^(A) Clinical Illness: a mosquito-borne viral illness characterized by acute onset and constitutional symptoms followed by a brief remission and recurrence of fever, hepatitis, albuminuria, and in some instances, renal failure, shock and generalized hemorrhages.

^(B) Refer to the [National Microbiology Laboratory Guide to Services](#) for current specimen collection and submission information.

^(C) Testing for yellow fever IgM is sent via the Public Health Laboratory (ProvLab) to the Centers for Disease Control and Prevention (CDC) in the United States (US) by request.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- MOH (or designate) of the zone.

Alberta Health Services and First Nations Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Reference 1 applies to this section.

Yellow fever virus is a flavivirus. The virus is unstable in the environment and is sensitive to heat (60°C for 10 minutes), ultraviolet radiation, disinfectants (alcohol and iodine), and acid pH. The infectious dose is unknown.

Clinical Presentation

Yellow fever is an acute, infectious viral disease of short duration and varying severity. The disease ranges from a self-limited infection to hemorrhagic fever.

Clinically mild cases may be unapparent, accounting for 5–50% of infections. A typical attack is characterized by three phases beginning with the sudden onset of fever and chills, headache, and myalgias, along with facial flushing and bradycardia (Faget's sign). These symptoms correspond to the viremic phase of infection. In the majority of cases, the end of this period signals the end of the illness, but in others remission may occur. The remission (second phase) may be brief before moving into the final phase characterized by renewed symptoms (the toxic phase) of high fever, headache, backache, nausea, vomiting, abdominal pain, and somnolence. Hemorrhagic fever with bleeding in the gastro-intestinal tract, hematemesis, gum bleeding, and petechiae and purpura may follow. Jaundice may appear early and intensify later in the disease. Cardiovascular instability may occur along with albuminuria, oliguria, and myocarditis. Confusion, seizures, and coma typify the late stages of illness. Fifty per cent of cases are fatal, with death occurring with seven to 10 days after onset. Secondary bacterial infections are common.

Diagnosis

Severe yellow fever resembles several other viral hemorrhagic fevers that occur in Africa and South America. Laboratory diagnosis is critical in making the diagnosis. In addition the patient's clinical features, places and dates of travel, immunizations, and activities will assist in making the diagnosis.

Treatment

- There is no specific treatment for yellow fever.
- Symptomatic treatment.

Reservoir

Reference 2 applies to this section.

There are three transmission cycles: jungle (sylvatic), intermediate, and urban type. All three cycles exist in Africa, but in South America, only jungle and urban types occur.

Jungle yellow fever occurs in tropical rainforests where monkeys infected by mosquitoes pass the virus onto other mosquitoes that feed on them. These mosquitoes in turn bite and infect humans entering the forest. The *Haemagogus* mosquitoes in South America and the *Ae. africanus* in Africa acquire the disease from monkeys and possibly marsupials.

The intermediate cycle of yellow fever transmission occurs in humid or semi-humid savannahs of Africa and can produce small-scale epidemics in rural villages. Mosquitoes infect both monkey and man. This is the most common type of outbreak seen in Africa.

In urban yellow fever, humans serve as the viremic host and the disease is spread between humans by the *Aedes aegypti* mosquito. Urban yellow fever results in large epidemics when travellers from rural areas introduce the virus into areas with high human populations.

Transmission

Yellow fever is transmitted from person-to-person through the bite of a mosquito, typically a day-time biter. The mosquito becomes infected after feeding on a viremic human and then spreading the infection in subsequent blood meals,

The risk of epidemic transmission occurs when a person with a forest-acquired infection travels to an *Ae. Aegypti* infested location while viremic.

Incubation Period

The incubation period is typically three to six days. Persons may be viremic before they demonstrate symptoms.

Period of Communicability

The blood of the host is infective for the mosquito shortly before the onset of the fever and the initial three to five days of illness. Mosquitoes become infective nine to 12 days after feeding on a viremic host and remain so for life (generally one to four months). It is not communicable by contact or common vehicles.

Host Susceptibility

Recovery from infection is followed by lasting immunity. Antibodies appear in the blood within the first week. Infants born to immune mothers may demonstrate passive immunity for up to six months.

Incidence

General

Yellow fever disease was first recognized in an outbreak in 1648 in the New World. The virus was introduced by trading vessels infested with the *Ae. aegypti* mosquito. In 1793, an epidemic in Philadelphia saw 1/10 of the population die. In 1878, the Mississippi Valley experienced 100,000 cases. The method of spread via the mosquito was not recognized until 1900.

Jungle yellow fever is rare and occurs most often in individuals who work in tropical rain forests. Urban yellow fever is the cause of most yellow fever outbreaks.

Outbreaks continue to occur periodically in South America and areas of sub-Saharan Africa, which predominantly include the moist savannah zones of West and Central Africa during the rainy season, occasionally in urban locations and villages in Africa, and less often in jungle regions. In these countries the disease is endemic and intermittently epidemic. The disease has not been documented in Asia or, to date, in the southern US.

The incidence of yellow fever in South America is lower than that in Africa because virus transmission between monkeys and mosquitoes occurs isolated from human contact, and immunity in the indigenous human population is high. In South America, cases occur most frequently in young men who have occupational exposures to mosquito vectors in forests or transitional areas of Bolivia, Brazil, Columbia, Ecuador, Venezuela and Peru. In West Africa the most dangerous time of year is during the late rainy and early dry seasons (July to October). In Brazil, virus transmission is highest during the rainy season (January to March). From 1996 to 2002, five cases of yellow fever were diagnosed in the US and Europe combined. Four of the five cases were acquired in South America, and all were fatal. Based on US data, the risk for illness in a traveller due to yellow fever has been estimated to be 0.4 to 4.3 per million travellers to yellow fever endemic areas.

Canada⁽³⁾

There were no cases of yellow fever reported in Canada from 1924 to 2003.

Alberta⁽⁴⁾

No cases of yellow fever have been reported in Alberta.

Public Health Management

Key Investigation

- Determine history of recent living in, immigration from, or travel to Africa or South America including:
 - geographic location,
 - season,
 - duration of exposure, and
 - occupational and recreational activities (while travelling).
- Determine history of mosquito bites.
- Determine immunization status.

Management of a Case

- Supportive care.
- Routine practices for hospitalized individuals.

Management of Contacts

- There is no follow-up of contacts as the disease is not spread directly from person to person, but rather, person-mosquito-person.
- To find unreported or undiagnosed cases, it may be prudent to identify others who may have been in the same location in the two weeks prior to the case becoming ill.

Preventive Measures

- Travellers should be advised to confirm their need for yellow fever vaccine with the appropriate clinic before travelling and receiving the vaccine.
 - Most countries have regulations and requirements for vaccination that must be followed prior to entering the country.
 - Immunize with yellow fever vaccine prior to travel when appropriate.
 - The vaccine is almost 100% effective and lasts 10 years or more commencing 10 days after it has been administered.
 - The vaccine is administered at designated yellow fever vaccination centres.
 - AH does not fund yellow fever vaccine.
- Use mosquito precautions.
 - Remain in well screened or completely enclosed, air-conditioned areas when mosquitoes are active.
 - Wear loose fitting, light coloured clothing with full length pant legs and sleeves.
 - Use insect repellent (e.g., DEET containing) on exposed skin and clothing.
- Laboratory safety – Yellow fever requires biosafety level 3 practices.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	• Updated Template • Etiology, Clinical Presentation, Diagnosis and Treatment sections moved to Epidemiology • Key Investigation section moved to Public Health Management (formerly called Control) • Updated web links

References

1. Public Health Agency of Canada, *Infectious substances: Yellow fever virus*, Office of Laboratory Security, Material Safety Data Sheet, November 1999.
2. World Health Organization, *Epidemic and Pandemic Alert Response (EPR) – Yellow Fever*, 2005.
3. Public Health Agency of Canada, *Notifiable Diseases On-line – Yellow fever*, February 2004.
4. Alberta Health and Wellness, Disease Control and Prevention, *Communicable Disease Reporting System*, August 2002.